Asymptotic Normality of the LDSC Estimator Under Fixed Genetic Effects

1 Introduction and Setup

We consider the proportional asymptotic regime where $n, n_r, p \to \infty$ such that $p/n \to \gamma_1 \in (0, \infty)$ and $p/n_r \to \gamma_2 \in (0, \infty)$.

Assumption 1 (Distributions)**.** *The rows of* $\mathbf{X} \in \mathbb{R}^{n \times p}$ *and* $\mathbf{Z} \in \mathbb{R}^{n_r \times p}$ *are independent and identically distributed as* $N(0, \mathbf{\Sigma})$*. The environmental noise is* $\boldsymbol{\epsilon} \sim N(0, \sigma_\epsilon^2 \mathbf{I}_n)$*, independent of* $\mathbf{X}$ *and* $\mathbf{Z}$*.*

Assumption 2 (Covariance Structure)**.** *The true covariance matrix* $\mathbf{\Sigma}$ *is block-diagonal with* K *blocks* $\mathcal{B}_1, \ldots, \mathcal{B}_K$*. The maximum block size is bounded by a constant* $M < \infty$*, and the eigenvalues of* $\mathbf{\Sigma}$ *are strictly bounded away from* 0 *and* ∞*.*

Assumption 3 (Fixed Genetic Effects)**.** *The genetic effects* $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top \in \mathbb{R}^p$ *are fixed, deterministic constants. To ensure a non-degenerate asymptotic limit, we assume the effects are uniformly bounded in scale such that* $\max_{1 \le j \le p} |\alpha_j| = O(p^{-1/2})$ *and* $\lim_{p \to \infty} \|\boldsymbol{\alpha}\|_2^2 = \tau^2 > 0$*.*

2 Linearization of the Estimator

The LDSC estimator is defined as:

$$\hat{\sigma}_\alpha^2 = \frac{\frac{1}{p}(\hat{\ell} - \mu_{\hat{\ell}}\mathbf{1})^\top (\hat{w} - \hat{\mu}_w \mathbf{1})}{\frac{1}{p}(\hat{\ell} - \mu_{\hat{\ell}}\mathbf{1})^\top (\hat{\ell} - \mu_{\hat{\ell}}\mathbf{1})} = \frac{\mathcal{N}_p}{\mathcal{D}_p}.$$

First, we analyze the denominator $\mathcal{D}_p$. Because $\hat{\ell}_j = \sum_{i=1}^p \widehat{\mathbf{\Sigma}}_{ji}^2$ where $\widehat{\mathbf{\Sigma}} = \frac{1}{n_r}\mathbf{Z}^\top \mathbf{Z}$, the sequence $\{\hat{\ell}_j\}_{j=1}^p$ is strictly a function of the reference panel $\mathbf{Z}$. Due to the block-diagonal structure of $\mathbf{\Sigma}$, $\hat{\ell}_j$ and $\hat{\ell}_k$ are asymptotically independent if they belong to different blocks. By the Law of Large Numbers for weakly dependent sequences, $\mathcal{D}_p$ converges in probability to a strictly positive constant:

$$\mathcal{D}_p \xrightarrow{P} \lim_{p \to \infty} E[\mathcal{D}_p] = d_0 > 0.$$

By Slutsky's Theorem and the Delta method, the asymptotic normality of $\hat{\sigma}_\alpha^2$ is governed by the joint asymptotic normality of the numerator and denominator. We define the linearized statistic:

$$g(\mathbf{X}, \mathbf{Z}, \boldsymbol{\epsilon}) = \mathcal{N}_p - \theta \mathcal{D}_p,$$

where $\theta = \lim_{p \to \infty} E[\mathcal{N}_p]/E[\mathcal{D}_p]$. If we show that $\sqrt{p}(g - E[g]) \xrightarrow{d} N(0, V^2)$, it follows that $\sqrt{p}(\hat{\sigma}_\alpha^2 - \theta) \xrightarrow{d} N(0, V^2/d_0^2)$.

3 Variance Scaling via Moment Calculations

To apply Chatterjee's theorem[cite: 56], we establish that the variance of the unscaled statistic g is of order $O(p^{-1})$. Let $\mathbf{v} = (\mathrm{vec}(\tilde{\mathbf{X}}), \mathrm{vec}(\tilde{\mathbf{Z}}), \tilde{\boldsymbol{\epsilon}})$ be the standard normal vector such that $\mathbf{X} = \tilde{\mathbf{X}}\mathbf{\Sigma}^{1/2}$, $\mathbf{Z} = \tilde{\mathbf{Z}}\mathbf{\Sigma}^{1/2}$, and $\boldsymbol{\epsilon} = \sigma_\epsilon \tilde{\boldsymbol{\epsilon}}$.

The term $\hat{w}_j = \frac{1}{n^2}(\mathbf{X}_j^\top \mathbf{X}\boldsymbol{\alpha} + \mathbf{X}_j^\top \boldsymbol{\epsilon})^2$ is a quartic form in Gaussian variables. Utilizing the exact Wishart moment formulations derived in Proposition S1 of Dicker (2014)[cite: 193, 218], we can evaluate the variance. Because $\boldsymbol{\Sigma}$ is block-diagonal with maximum block size M, $\text{Cov}((\hat{\ell}_j - \mu_{\hat{\ell}})\hat{w}_j, (\hat{\ell}_k - \mu_{\hat{\ell}})\hat{w}_k) = 0$ for variants j, k in different blocks. Thus, the double sum expansion of $\text{Var}(g(\mathbf{v}))$ contains at most pM non-zero terms. Since each term is $O(1)$ under the bounded fixed effects scaling, we obtain:

$$\psi^2 = \text{Var}(g(\mathbf{v})) = \Theta(p^{-1}).$$

4 Asymptotic Normality via Chatterjee's Theorem

We apply Theorem 2.2 from Chatterjee (2009) (restated as Theorem S1 in Dicker 2014)[cite: 56, 57]. For $\mathbf{v} \sim N(0, \mathbf{I})$, let $W = (g(\mathbf{v}) - E[g(\mathbf{v})])/\psi$ and $Z \sim N(0, 1)$. The total variation distance is bounded by:

$$d_{TV}(W, Z) \leq \frac{2\sqrt{5}\kappa_1\kappa_2}{\psi^2},$$

where $\kappa_1 = E[\|\nabla g(\mathbf{v})\|^4]^{1/4}$ and $\kappa_2 = E[\|\nabla^2 g(\mathbf{v})\|_{\text{op}}^4]^{1/4}$[cite: 59].

1. Bounding the Gradient (κ_1):
We compute the partial derivatives of g. For $x_{ik} \in \tilde{\mathbf{X}}$:

$$\frac{\partial g}{\partial x_{ik}} = \frac{1}{p}\sum_{j=1}^{p}(\hat{\ell}_j - \mu_{\hat{\ell}})\frac{\partial \hat{w}_j}{\partial x_{ik}}.$$

Because $\frac{\partial \hat{w}_j}{\partial x_{ik}}$ scales as $O(n^{-1})$ (following analogous bounding techniques to Dicker's Lemma S1 [cite: 15, 76]), and the sum aggregates over localized blocks, the squared norm aggregated over all $n \times p$ entries yields $\|\nabla_x g\|^2 = O_p(n^{-1})$. Similar bounds apply to $\nabla_z g$ and $\nabla_\epsilon g$. Hence, $\kappa_1 = O(p^{-1/4})$[cite: 104, 106].

2. Bounding the Hessian (κ_2):
The Hessian $\nabla^2 g(\mathbf{v})$ is block-sparse due to the local dependence structure. The second derivatives $\frac{\partial^2 g}{\partial x_{ik}\partial x_{i'k'}}$ are $O(n^{-2})$. Bounding the operator norm of this localized block matrix (following Dicker's methodology for equation S11 [cite: 112, 141]) demonstrates that it scales strictly inversely with the dimension: $\|\nabla^2 g(\mathbf{v})\|_{\text{op}} = O_p(p^{-1})$. Thus, $\kappa_2 = O(p^{-1})$[cite: 145, 147].

3. Synthesis:
Substituting κ_1, κ_2, and ψ^2 into Chatterjee's inequality:

$$d_{TV}(W, Z) \leq C\frac{p^{-1/4} \cdot p^{-1}}{p^{-1}} = O(p^{-1/4}).$$

As $n, p \to \infty$, the total variation distance vanishes: $O(p^{-1/4}) \to 0$[cite: 63, 149].

5 Conclusion

Since $d_{TV}(W, Z) \to 0$, the standardized statistic converges to a standard normal distribution:

$$\frac{g(\mathbf{v}) - E[g(\mathbf{v})]}{\sqrt{\text{Var}(g(\mathbf{v}))}} \xrightarrow{d} N(0, 1).$$

By the continuous mapping theorem and Slutsky's theorem established in Section 2, we conclude that the centered LDSC estimator under fixed effects is asymptotically normal:

$$\frac{\hat{\sigma}_\alpha^2 - E[\hat{\sigma}_\alpha^2]}{\sqrt{\text{Var}(\hat{\sigma}_\alpha^2)}} \xrightarrow{d} N(0, 1).$$